

## Area 1: Grok Code + Vision + Computer Use for mobile app developers

### 1) Why are you interested in this area?

Mobile app development is a massive industry wherein code generation LLMs are still lacking, a problem which I've faced intimately when I attempt to build responsive Kotlin or React Native apps using LLM code assistance. When using LLM code, I have many frustrations with poor UI appearance, unintended UX gestures, and *especially* broken permissions and integrations with native features such as camera, notifications, timers, etc.

My frustrating experience has also given me insight into what needs to be solved:

- good mobile-native code (Swift/Kotlin) examples are much rarer than say, React
- There's no strong end-to-end testing harness an LLM can use to write → test on a phone with complicated UX gestures and permissions with computer use models → analyze with vision → optionally take dev input → iterate.

I also think the time is perfect to tackle mobile development because LLMs have had meteoric growth in their ability to write web development code, which is honestly a similar problem with just more data and harnesses – so this problem is likely tractable. No frontier lab is clearly leading here, and there is opportunity since purpose-built community benchmarks (Kotlin-bench V2 from Firebender, Android Bench from Google) launched in early 2026 specifically to address this.

Regarding economics, there is still no clear winner for mobile code writing and it is a massive market that wants to be captured.

### 2) How good do you think grok is in this area and why?

While there haven't been many evals done with Grok on mobile code generation tasks, the general landscape is brutal even with top code models such as Claude Sonnet, so I would expect Grok's performance to be similar:

MobileDev-Bench (March 2026, 384 real-world issue-resolution tasks across 18 production mobile apps spanning Android Native Java/Kotlin, React Native TypeScript, and Flutter Dart): "Evaluation of four state-of-the-art code-capable LLMs, GPT-5.2, Claude Sonnet 4.5, Gemini Flash 2.5, and Qwen3-Coder, yields low end-to-end resolution rates of 3.39%–5.21%, revealing significant performance gaps compared to prior benchmarks." For context, the same caliber of models hit 77–85% on SWE-bench Verified (web/Python repo benchmark). So we're looking at roughly a 15–20× drop in resolution rate when frontier code models move from web to mobile. [<https://arxiv.org/abs/2603.24946>]

xAI has not published mobile-, Kotlin-, or Swift-specific evaluations for any Grok version, but neither have most frontier labs. I think this is a telling gap and a real opportunity to get ahead of the curve.

### 3) Share a Grok conversation you have to assess what could be good evals for this specific area and how to approach building evals for this area

In this conversation, I ask Grok 4.3 to develop a Kotlin geocaching app that heavily uses location and compass APIs. I noticed many mistakes and omissions in the code in the first response, and there were several aspects relating to versions, components, and permissions it backtracked or seemed unconfident on. [https://grok.com/share/bGVnYWN5\\_0b90c6b6-373e-41ee-92e5-d59744f9dc58](https://grok.com/share/bGVnYWN5_0b90c6b6-373e-41ee-92e5-d59744f9dc58)

It's worth noting that I had given Grok 4.3 the exact same prompt but asked it to make it in React Native, which it did in one shot, correctly, without any omissions, showcasing the capability gap. Still, there were minor UI bugs in this app I wanted to fix.

Evaluation ideas:

1. Fast upskilling with public benchmarks. Android Bench (Google), MobileDev-Bench, Kotlin-bench V2, SwiftEval are all missing Grok Code evaluations.
2. Vision + Computer Use Harness. We develop a Grok vision + computer-use-powered harness to enable Grok to continuously iterate on mobile development tasks by coding, running it on a phone/phone emulator, and using computer use for gestures and vision for verification. We can use this harness and evaluate the number of iterations it takes for Grok to complete tasks and whether it is successful in the end. When xAI releases Grok Build, this could be its own feature too.

#### **4) 3 specific X profiles of people that could represent power users and be great to guide you on shaping evals/giving feedback**

Matt Gallagher ([@cocoawithlove](#)): popular mobile software dev on X

"Have LLMs improved for Swift coding in the last 12 months? ... All these LLMs feel completely random. Change a couple words and they'll give completely different results — often going from working to not-working and back again. If you don't like unpredictability, it's still not a comfortable approach to programming."

Andrey Breslav ([@abreslav](#)): Creator of Kotlin, founder of CodeSpeak.

Authority on the language, with measured-skeptic public stance on AI coding tools.

Krzysztof Zablocki ([@merowing](#)): Principal Engineer at The Browser Company, Author of open-source tools like Sourcery, Inject, and more.

"There's significantly less Swift code in the training data compared to JavaScript or Python. What you get is code that compiles, but doesn't follow the patterns you use in production: no dependency injection, missing your testing approach, ignoring your architecture conventions. The gap between tutorial code and production/commercial code is massive."

## Area 2: Grok chat to support physical, chemical, materials scientists in reasoning, literature analysis, and hypothesis evaluation

### 1) Why are you interested in this area?

Producing models with strong capabilities in reasoning and epistemically-grounded creativity in the physical sciences can massively accelerate scientific discovery. This is the dream. This is still a very open problem and right in line with xAI's mission to understand the universe.

While there are labs (i.e. FutureHouse, etc.) tackling this, I think the unfair adoption potential of Grok and how strongly it performs as a closed-source model *that beats out other frontier labs on hallucination* gives us a head start and makes this problem much more feasible.

Regarding economics, there is a large, open market with mature physical sciences industries and academia labs. The Grok brand and strong performance in this area could be an avenue toward embedding Grok into many more enterprises.

### 2) How good do you think grok is in this area and why?

I believe that while Grok performs well in aggregate intelligence scores, benchmarks and online opinion paint a picture that it is weak in physical sciences' epistemically-grounded reasoning. *Correction: after some further Grok exploration in Q3, I think Grok 4.3 is quite strong in this area and I think some re-evaluations of benchmarks mentioned here with Grok 4.3 are in order!*

The pros:

- Grok 4.3 scores 53 on the Artificial Analysis Intelligence Index, well above the median of 36 for reasoning models in its price tier.
- **Grok 4.20 0309 v2 (Reasoning) scores the lowest on AA-Omniscience Hallucination Rate with a score of 17%, followed by Grok 4.20 0309 (Reasoning) with a score of 22%, and MiMo-V2.5-Pro with a score of 25%. <https://artificialanalysis.ai/evaluations/omniscience>**

However,

- Working mathematicians evaluating Grok 4 found it "especially weak in algebraic combinatorics," that it "did not seem to understand the level of detail required by a working mathematician for a proof," that it "prefers very short answers" and even when coaxed its reasoning is "often full of gaps." It would "handwave its way through proofs that required rigorous but 'soft' arguments." [<https://epoch.ai/blog/grok-4-math>]
- On CMPhysBench (condensed matter physics, expert-built), Grok 4 is literally the best model, and it still gets crushed. "Our results show that even the best models, Grok-4, reach only 36 average SEED score and 28% accuracy on CMPhysBench, underscoring a significant capability gap, especially for this practical and frontier domain relative to traditional physics."
- CritPt evaluates language models on solving unpublished, frontier-level physics problems that require genuine research-scale reasoning. The best is GPT-5.5 Pro at 30.6%, while Grok 4.3 achieves only 8.0% [<https://artificialanalysis.ai/evaluations/critpt>]

### 3) Share a Grok conversation you have to assess what could be good evals for this specific area and how to approach building evals for this area

In this conversation, I provided Grok 4.3 with five quantum science-related hypotheses and asked it for an evaluation of their defensibility. Two of them actually appeared in a UCSB lab's papers' conclusions within the last two years, while the rest were shakier made-up ones that just sounded plausible.

[https://grok.com/share/bGVnYWN5\\_f086c25c-413f-4239-9edb-cacbfd21d35e](https://grok.com/share/bGVnYWN5_f086c25c-413f-4239-9edb-cacbfd21d35e)

I came in expecting underperformance based on my research in Q2 – Epoch's math review, CMPPhysBench at 28%, CritPt at 8%. However, Grok 4.3 crushed this. I'm very excited to get Grok respected and used in actual physical sciences labs by running evals with more breadth. And plus, the benchmark data indicates even Grok 4.3 has significant gaps, and I'm excited to address them with robust evals.

Evaluation ideas:

1. Difficult physical science reasoning benchmarks. Benchmarks such as ChemBench, MatSciBench, MaCBench use graduate-level physical sciences problems that might be asked in a research lab with ready-to-use eval infrastructure: public datasets, automated grading pipelines, and SEED-style partial-credit scoring. We could have an RL loop on top of these.
2. Hypothesis differentiation evaluation assisted by experts. We produce an evaluation dataset of strong and weak hypotheses from physical sciences research papers that don't exist in training data, perhaps using peer review data and expert annotations. We evaluate the model's ability to concisely and correctly differentiate the strong hypotheses, and produce well-calibrated/consistent results. With expert feedback, we can create an RLHF loop.

### 4) 3 specific X profiles of people that could represent power users and be great to guide you on shaping evals/giving feedback

Lee Cronin ([@leecronin](#)): Regius Professor of Chemistry at the University of Glasgow

"Probabilistic slop engines cannot do science, drug discovery, materials discovery... Anyone who thinks AI can autonomously do science simply doesn't understand how knowledge is created."

"The biggest category error in AI for science is viewing a science problem as a search problem..."

Jonathan Oppenheim ([@postquantum](#)): Professor of Physics at University College London

"This is what I mean by science slop: work that looks plausibly correct and technically competent but isn't, and doesn't advance our understanding. It has the \*form\* of scholarship without the \*substance\*. The formalism looks correct, the references are in order, and it will sit in the literature forever, making it marginally harder to find the papers that actually matter."

Markus J. Buehler ([@ProfBuehlerMIT](#)): McAfee Professor of Engineering at MIT

"Even the most advanced LLMs/reasoning models collapse facts and beliefs into one surface. Until AI learns to differentiate, it cannot genuinely discover - science is, after all, the art of differentiation. At the frontier we need systems that learn how to represent, revise, and reconcile belief states through compositional, interpretable reasoning."

### Area 3: Making Grok Voice + Chat much more pedagogically effective for senior or graduate students in STEM studies.

#### 1) Why are you interested in this area?

As a college student studying upper division CS + Math and someone constantly trying to keep up with the frontier, from agentic tools to AI architecture research papers, I've hit the same wall over and over when using AI to learn advanced topics:

1. It takes too many conversation turns to get to the actually-good explanations. Models default to wall-of-text info-dumps instead of lecturer-paced step-by-step.
2. I have to keep reminding the model to speak in concise, high-signal sentences. The bias toward verbosity is so strong.
3. Voice has insanely high potential as a faster teaching medium and an easier way to voice confusion, but it's largely ineffective right now because the model literally dictates LaTeX characters (arrow, dollar, equal sign) instead of explaining the math, ideally paired with a visual.

I believe the model is capable of being a strong teacher since I do get good explanations in the end. The real problem is getting there faster and biasing toward shorter, lecturer-style answers — which is a tractable post-training problem.

#### 2) How good do you think grok is in this area and why?

I believe Grok is relatively poor compared to other frontier models in this area.

1. Grok's consistently missing from different pedagogy benchmarks.  
<https://eth-lre.github.io/mathtutorbench/>, MMTutorBench, MRBench, KMP-Bench, SafeTutors, SocraticLM also don't mention it.
2. Grok scored very well on different math exams and has been highly optimised for problem solving, but "specializing an LLM for pedagogy comes at the cost of solving ability and, conversely, a high solving accuracy often means that the LLM lacks pedagogy."  
[\[https://arxiv.org/abs/2502.18940\]](https://arxiv.org/abs/2502.18940) Without any specific conditioning to be more socratic-like or use teacher phrasing in learning contexts, I doubt high performance could be expected.

If I may add my own experience, I find Grok to be poor in pedagogy as its text explanations are not high-signal enough for my own studies (currently studying neural networks + parallel processing + physics for example). Usually Claude Opus does the job for me, but even that is difficult to work with. Plus, the voice mode reduces to little more than a TTS engine over math and formulae which isn't helpful, just noisy.

#### 3) Share a Grok conversation you have to assess what could be good evals for this specific area and how to approach building evals for this area

A recent conversation I had with Grok to explain some MPI parallelism concepts:

[https://grok.com/share/bGVnYWN5\\_f2fa3149-cfc6-470f-835a-5d6e31596a24](https://grok.com/share/bGVnYWN5_f2fa3149-cfc6-470f-835a-5d6e31596a24)

The specific failure modes in teaching I encountered in this were:

1. Inaccurate understanding of subject material (which I only realized after switching to another model to teach me)
2. Info-dumping without self-verification, pedagogical language, or even consistency between responses.
3. Voice mode was literally dictating mathematical formulae and symbols, such as arrows, making it difficult to listen and learn from.

In short, it couldn't teach me one slide even after four turns of conversations.

Evaluation ideas:

1. Biasing toward teacher-shaped responses. This class of evaluation involves measuring how often Grok uses pedagogical-style utterances in responses via several benchmarks mentioned in question 2. For example, asking guiding questions instead of stating answers, hinting at next steps without giving them away, naming misconceptions without correcting it directly, building on what the student just said, and scaffolding incrementally instead of just revealing answers or info-dumping. It's been shown in research that just improving the pedagogical phrasing of LLMs tangibly improves an LLM's ability to teach human students! "The hierarchy of effects was clear: LLMs framed as interactive tutors ( $g=0.842$ ) generated substantially larger improvements in performance than when they functioned as passive tools ( $g=0.436$ )."  
[<https://arxiv.org/pdf/2509.22725>]
2. Directly testing pedagogical effectiveness in specific learning settings. This class of evaluation involves either (1) simulating students with LLMs or (2) using real life students and injecting Grok into the learning process to see if learning became more effective. For (2), real-life students, we could look to aid those studying specific exams such as AMC/AIME whose nature allows for well-scoped, bite-sized learning tasks, and which Grok itself already does well on. We measure the number of conversation turns and token usage until a student sends a binary "fully understood" signal for a problem proof. For (1), LLMs, we can have a pedagogically-improved Grok attempt to teach a smaller model to successfully complete highly reasoning-heavy tasks such as those from ARC-AGI 3. We measure the smaller model's success rate after pure textual "coaching" from Grok and number of conversation turns used.

#### **4) 3 specific X profiles of people that could represent power users and be great to guide you on shaping evals/giving feedback**

Dan Meyer ([@ddmeyer](#)): Former high school math teacher + Desmos Chief Academic Officer + leads AI feature dev at Amplify

"I came to the conclusion, which I'm still working on, that [AI] doesn't meet the core needs of teachers and students."

Sal Khan ([@salkhanacademy](#)): obvious pick

Khanmigo, Khan Academy's internal AI chatbot for teaching, has received very strong reviews and uses a socratic style of teaching. However, it is gated to Khan Academy courses and simpler subjects (K-12).

Justin Reich ([@bjfr](#)): Director Of Teaching Systems Lab & Associate Professor @ MIT

"My career has been devoted to studying education technology. Over and over again, we've seen new technologies emerge in education, and the technology developers will promise, every time, that the new tech will transform and democratize education."